

31338 Network Servers

Lab 3b Establishing and maintaining time and date settings

Aims

1. To set the correct timezone and set the current time in Linux using an NTP server
2. To set up an NTP daemon so that the correct time is always maintained

Task 1: Set the timezone

Use the `date` command to identify the current UTC and local times (using the `-u` option for UTC, and no option for the local timezone). If the local time zone is incorrectly identified, change `/etc/localtime` into a symbolic link to the correct file for our timezone under the directory `/usr/share/zoneinfo/`

For experience, try setting your system to a different timezone to verify that you can. When done, reset it to Australia/Sydney.

Task 2: Maintain correct time with chrony

In the latest versions of Centos, chrony is the system that maintains network time. Older systems use ntpd. Chrony consists of both a server (daemon) and a client, called `chronyd` and `chronyc` respectively.

This task is a matter of setting up the `chronyd` daemon to run when the machine boots, to maintain the correct system time. First, see whether `chronyd` is already running using the `ps` command (`ps -ef | grep chronyd`). Then see whether `chronyd` is known by `systemctl` (`systemctl status chronyd`).

Use `systemctl` to **start** `chronyd` running now. Also use `systemctl` to **enable** `chronyd` so that it will start when the machine boots.

To verify that chrony is working, use the client program (`chronyc`) as follows. Document these commands in your journal.

- `chronyc sources`
- `chronyc tracking`
- `chronyc add server time.uts.edu.au` (check the result with the “sources” command).

To test that it will synchronise, let's change the system date. Let's party like it's New Years Eve 1999:

- `date 123123591999.00` (look up the manual for the `date` command to understand this format)

Then just run `date` with no arguments to check the date. Does it reset to the current date/time? Why or why not? (Hint: use Google to learn more about how Chrony works)

Restart `chronyd`, using `systemctl`. After restarting the daemon, run the `date` command a few times. What happens now? Why?

Make notes in your journal about what happens and why. Also look at the configuration file `/etc/chrony.conf`

Task 3: Windows NTP

Windows can also automatically update the time and date via NTP. In Server Manager / Local Server, click on the time zone shown to open the Date and Time window (or find it through Control Panel). Go to the Internet Time tab.

You can then change the settings to choose your internet time server. The default is `time.microsoft.com` – try changing this to: `2.pool.ntp.org`